

INFINITELY MANY INTEGER SOLUTIONS OF $y^2 + x^2y + z^2x + 3 = 0$ VIA A PELL FIBER

CLAUDE (ANTHROPIC)

ABSTRACT. We prove that the Diophantine equation $y^2 + x^2y + z^2x + 3 = 0$ has infinitely many solutions in integers. The proof is fully explicit: on the fiber $x = -3$ the equation is equivalent to the Pell-type conic $s^2 - 12z^2 = 69$ with $s = 2y + 9$, which carries the lattice point $(9, 1)$ and is preserved by the automorphism induced by the fundamental unit $7 + 4\sqrt{3}$ of $\mathbb{Z}[\sqrt{3}]$. This produces an explicit exponentially growing sequence of solutions, with closed form $s_n + 2z_n\sqrt{3} = (9 + 2\sqrt{3})(7 + 4\sqrt{3})^n$. We also explain the structural reduction that locates this fiber, and contrast it with the degenerate fiber $x = -1$, on which the conic splits over $\mathbb{Z}$ and admits exactly four solutions.

1. INTRODUCTION AND MAIN RESULT

We study the integer solutions of

$$(1) \quad F(x, y, z) := y^2 + x^2y + z^2x + 3 = 0, \quad (x, y, z) \in \mathbb{Z}^3.$$

Theorem 1.1. *Equation (1) has infinitely many integer solutions. Explicitly, define $(y_n, z_n)_{n \geq 0}$ by*

$$(y_0, z_0) = (0, 1), \quad y_{n+1} = 7y_n + 12z_n + 27, \quad z_{n+1} = 4y_n + 7z_n + 18.$$

Then for every $n \geq 0$ the triple $(x, y, z) = (-3, y_n, z_n)$ satisfies (1), the integers $z_0 < z_1 < z_2 < \dots$ are strictly increasing, and consequently the triples $(-3, y_n, \pm z_n)$ form an infinite set of pairwise distinct solutions. The first members are

$$(-3, 0, 1), \quad (-3, 39, 25), \quad (-3, 600, 349), \quad (-3, 8415, 4861), \quad (-3, 117264, 67705), \quad \dots$$

The proof occupies Sections 3–4. Section 2 records the reasoning that *finds* the family; Section 5 collects the symmetries of the solution set and the contrasting finite fiber $x = -1$.

2. INTUITION: WHY FIX x , AND WHY $x = -3$

Three observations organize the problem; all are made rigorous below.

(i) *The variable x is arithmetically rigid.* Reducing (1) modulo x kills the terms x^2y and z^2x and leaves $y^2 + 3 \equiv 0 \pmod{x}$ (Lemma 3.2). Thus x acts as a modulus constraining the other variables, which suggests *fixing* x and analyzing the resulting surface fiber by fiber.

(ii) *Each fiber is a conic, and generically a Pell conic.* For fixed $x = -c$, equation (1) is a quadratic in (y, z) . Completing the square turns it into $s^2 - 4cz^2 = c^4 - 12$ with $s = 2y + c^2$ (Lemma 3.3). Whenever $4c$ is *not* a perfect square, the binary form $s^2 - 4cz^2$ has irrational asymptotes and a nontrivial automorphism group coming from the Pell equation $u^2 - 4cv^2 = 1$; a single lattice point then propagates to infinitely many. When $4c$ *is* a square the form splits over $\mathbb{Z}$ and the fiber is finite; the fiber $c = 1$ exhibits exactly this degeneration (Proposition 5.2).

(iii) *A seed is handed to us by the slice $y = 0$.* Setting $y = 0$ in (1) collapses it to $z^2x = -3$, whose only solution with z^2 a square is $(x, z) = (-3, \pm 1)$. This simultaneously selects the fiber

$c = 3$ (for which $4c = 12$ is a nonsquare, so the Pell mechanism is available) and supplies the seed point $(s, z) = (9, 1)$ on the conic $s^2 - 12z^2 = 69$. The rest is the Pell machine.

3. PRELIMINARY REDUCTIONS

Lemma 3.1. *Equation (1) has no solution with $x = 0$.*

Proof. If $x = 0$ then (1) reads $y^2 + 3 = 0$, impossible since $y^2 \geq 0$. $\square$

Lemma 3.2. *If (x, y, z) solves (1), then $x \mid y^2 + 3$.*

Proof. By Lemma 3.1, $x \neq 0$. From (1), $y^2 + 3 = -x^2y - z^2x = x(-xy - z^2)$, and $-xy - z^2 \in \mathbb{Z}$. $\square$

Lemma 3.2 is not used in the proof of Theorem 1.1; it is the structural fact motivating the fiberwise analysis of Section 2(i).

Lemma 3.3 (Conic normal form of a fiber). *Let $c \in \mathbb{Z}$, $c \neq 0$, and put $x = -c$. Then for all $y, z \in \mathbb{Z}$,*

$$F(-c, y, z) = 0 \iff (2y + c^2)^2 - 4cz^2 = c^4 - 12.$$

Moreover the map $y \mapsto s = 2y + c^2$ is a bijection from $\mathbb{Z}$ onto the set of integers congruent to $c^2 \pmod{2}$, with inverse $y = (s - c^2)/2$.

Proof. With $x = -c$, equation (1) reads $y^2 + c^2y - cz^2 + 3 = 0$. Multiplying by 4 and completing the square in y ,

$$4y^2 + 4c^2y - 4cz^2 + 12 = (2y + c^2)^2 - c^4 - 4cz^2 + 12,$$

so $F(-c, y, z) = 0$ iff $(2y + c^2)^2 - 4cz^2 = c^4 - 12$; multiplication by the nonzero integer 4 changes no solution set. The final assertion is immediate. $\square$

Note the equivalence in Lemma 3.3 is *exact* in both directions: every lattice point (s, z) on the conic with $s \equiv c^2 \pmod{2}$ yields a solution of (1), and conversely.

4. THE PELL FIBER $x = -3$

Specializing Lemma 3.3 to $c = 3$:

$$(2) \quad F(-3, y, z) = 0 \iff s^2 - 12z^2 = 69, \quad s = 2y + 9.$$

Lemma 4.1 (Automatic parity). *Every $(s, z) \in \mathbb{Z}^2$ with $s^2 - 12z^2 = 69$ has s odd; hence $y = (s - 9)/2 \in \mathbb{Z}$, and every lattice point of the conic (2) gives an integer solution $(-3, \frac{s-9}{2}, z)$ of (1).*

Proof. $s^2 = 69 + 12z^2$ is odd, so s is odd; then $s - 9$ is even. The last claim follows from Lemma 3.3 with $c = 3$, since $s \equiv 1 \equiv 9 \pmod{2}$. $\square$

Lemma 4.2 (Pell automorphism). *Define $T : \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$ by $T(s, z) = (7s + 24z, 2s + 7z)$. Then*

$$(7s + 24z)^2 - 12(2s + 7z)^2 = s^2 - 12z^2 \quad \text{for all } s, z \in \mathbb{Z}.$$

In particular T maps the solution set of $s^2 - 12z^2 = 69$ into itself. Moreover, if $s > 0$ and $z > 0$ then $T(s, z) = (s', z')$ has $s' > s > 0$ and $z' > z > 0$.

Proof. Expanding,

$$(7s + 24z)^2 - 12(2s + 7z)^2 = (49 - 48)s^2 + (336 - 336)sz + (576 - 588)z^2 = s^2 - 12z^2.$$

Conceptually, T is multiplication by the unit $\varepsilon = 7 + 4\sqrt{3} \in \mathbb{Z}[\sqrt{3}]$, of norm $7^2 - 3 \cdot 4^2 = 1$, on elements written as $s + 2z\sqrt{3}$:

$$(7 + 4\sqrt{3})(s + 2z\sqrt{3}) = (7s + 24z) + 2(2s + 7z)\sqrt{3},$$

and $N(s + 2z\sqrt{3}) = s^2 - 12z^2$ is multiplicative, $N(\varepsilon\alpha) = N(\varepsilon)N(\alpha) = N(\alpha)$. For positivity: if $s, z > 0$ then $s' = 7s + 24z > s$ and $z' = 2s + 7z > z$ termwise. $\square$

Proof of Theorem 1.1. Set $s_n = 2y_n + 9$. We first check that the stated recurrence for (y_n, z_n) is T in the coordinates (s, z) :

$$s_{n+1} = 2y_{n+1} + 9 = 2(7y_n + 12z_n + 27) + 9 = 7(2y_n + 9) + 24z_n = 7s_n + 24z_n,$$

$$z_{n+1} = 4y_n + 7z_n + 18 = 2(2y_n + 9) + 7z_n = 2s_n + 7z_n,$$

so $(s_{n+1}, z_{n+1}) = T(s_n, z_n)$. The seed satisfies $(s_0, z_0) = (9, 1)$ and $9^2 - 12 \cdot 1^2 = 81 - 12 = 69$. By Lemma 4.2 and induction, $s_n^2 - 12z_n^2 = 69$ for all $n \geq 0$, and since $s_0, z_0 > 0$, also $s_n, z_n > 0$ with $z_{n+1} > z_n$ strictly. Both recurrences have integer coefficients and integer seed, so $y_n, z_n \in \mathbb{Z}$ for all n (alternatively, $y_n \in \mathbb{Z}$ by Lemma 4.1).

By the equivalence (2) (Lemma 3.3 with $c = 3$, together with Lemma 4.1), each $(-3, y_n, z_n)$ solves (1). Since z^2 alone appears, $(-3, y_n, -z_n)$ solves it as well. The z_n are strictly increasing, hence pairwise distinct, so the triples are pairwise distinct and the solution set of (1) is infinite. $\square$

Corollary 4.3 (Closed form and growth). *With $\varepsilon = 7 + 4\sqrt{3}$ and $\bar{\varepsilon} = 7 - 4\sqrt{3} = \varepsilon^{-1}$,*

$$s_n + 2z_n\sqrt{3} = (9 + 2\sqrt{3})\varepsilon^n \quad (n \geq 0),$$

equivalently

$$s_n = \frac{(9 + 2\sqrt{3})\varepsilon^n + (9 - 2\sqrt{3})\bar{\varepsilon}^n}{2}, \quad z_n = \frac{(9 + 2\sqrt{3})\varepsilon^n - (9 - 2\sqrt{3})\bar{\varepsilon}^n}{4\sqrt{3}},$$

and $y_n = (s_n - 9)/2$. In particular $z_n \sim \frac{9 + 2\sqrt{3}}{4\sqrt{3}}(7 + 4\sqrt{3})^n$ as $n \rightarrow \infty$.

Proof. Induction on n . For $n = 0$, $s_0 + 2z_0\sqrt{3} = 9 + 2\sqrt{3}$. For the step, by the displayed identity in the proof of Lemma 4.2, $\varepsilon(s_n + 2z_n\sqrt{3}) = s_{n+1} + 2z_{n+1}\sqrt{3}$. The explicit formulas follow by adding and subtracting the conjugate identity $s_n - 2z_n\sqrt{3} = (9 - 2\sqrt{3})\bar{\varepsilon}^n$ (obtained by applying the nontrivial automorphism of $\mathbb{Q}(\sqrt{3})$, which fixes the rational integers s_n, z_n). The asymptotic follows since $|\bar{\varepsilon}| < 1$. $\square$

TABLE 1. The first members of the family of Theorem 1.1 (each verified to satisfy $y^2 + 9y - 3z^2 + 3 = 0$).

n	y_n	z_n
0	0	1
1	39	25
2	600	349
3	8415	4861
4	117264	67705
5	1633335	943009

5. SYMMETRIES, SPORADIC SOLUTIONS, AND THE SPLIT FIBER $x = -1$

Remark 5.1 (Symmetries of the solution set). Since z enters (1) only through z^2 , the map $z \mapsto -z$ preserves solutions. On a fixed fiber $x = -c$ there is additionally the Vieta involution $y \mapsto -c^2 - y$ (i.e. $s \mapsto -s$), because y and $-c^2 - y$ are the two roots of the quadratic $Y^2 + c^2Y + (3 - cz^2)$. For $c = 3$ this sends the family of Theorem 1.1 to a second infinite family $(-3, -9 - y_n, \pm z_n)$, e.g. $(-3, -9, \pm 1)$ and $(-3, -48, \pm 25)$.

Proposition 5.2 (The fiber $x = -1$ is finite). *The solutions of (1) with $x = -1$ are exactly $(-1, 2, \pm 3)$ and $(-1, -3, \pm 3)$.*

Proof. By Lemma 3.3 with $c = 1$: $F(-1, y, z) = 0$ iff $s^2 - 4z^2 = -11$ with $s = 2y + 1$. Here $4c = 4$ is a perfect square and the form factors over $\mathbb{Z}$: $(s - 2z)(s + 2z) = -11$. Write $a = s - 2z$, $b = s + 2z$, so $ab = -11$ and, since 11 is prime, $(a, b) \in \{(1, -11), (-1, 11), (11, -1), (-11, 1)\}$. In each case $s = (a + b)/2$ and $z = (b - a)/4$ must be integers; all four cases pass, giving $(s, z) \in \{(-5, -3), (5, 3), (5, -3), (-5, 3)\}$, i.e. $s = \pm 5$, $z = \pm 3$. Then $y = (s - 1)/2 \in \{2, -3\}$, and one checks directly that all four resulting triples satisfy (1): $4 + 2 - 9 + 3 = 0$ and $9 - 3 - 9 + 3 = 0$. $\square$

Proposition 5.2 is the precise sense in which the Pell mechanism can fail: when $4c$ is a square the conic degenerates into a pair of rational lines and admits only the finitely many lattice points dictated by the divisors of $c^4 - 12$. The choice $c = 3$ in Theorem 1.1 is the smallest fiber that is both nondegenerate ($12 \notin \mathbb{Z}^2$) and *populated*, the seed being supplied by the slice $y = 0$ of (1): $y = 0$ forces $z^2x = -3$, whence $x = -3$, $z = \pm 1$ (the alternative $x = -1$, $z^2 = 3$ being impossible).

Remark 5.3 (Other sporadic solutions). The slice $z = 0$ also yields solutions, though only finitely many on that slice: (1) becomes $y(y + x^2) = -3$, so $y \mid 3$; checking $y \in \{\pm 1, \pm 3\}$ gives exactly $(\pm 2, -1, 0)$ and $(\pm 2, -3, 0)$. These lie on the fibers $x = \pm 2$ and are unrelated to the infinite family above; they illustrate that the full solution set of (1) is a union of fiberwise conic loci of varying richness. A complete classification of *all* solutions would require deciding, for every c with $4c$ nonsquare, whether the Pell-type equation $s^2 - 4cz^2 = c^4 - 12$ is soluble — a genuinely fiber-by-fiber question that we do not pursue, Theorem 1.1 already settling the dichotomy posed (“finitely many, listed, or provably infinitely many”) in favor of infinitude.